

PROJECT REPORT

Virtual Agent–Driven SLA Breach Awareness & Justification System

1. INTRODUCTION

1.1 Project Overview

Service Level Agreements (SLAs) are essential components of IT Service Management (ITSM) that define the agreed response and resolution times between service providers and customers. Meeting SLA commitments is crucial because it directly affects customer satisfaction, operational efficiency, and organizational reputation. In many organizations, however, SLA breaches continue to occur due to delayed responses, lack of proactive monitoring, insufficient communication between support teams, and absence of proper governance mechanisms. These issues often result in increased escalations, reduced productivity, compliance challenges, and dissatisfied customers.

Traditional SLA monitoring methods mainly depend on manual supervision by support teams and managers. As organizations grow and the number of incidents increases, manually tracking each SLA becomes difficult and error-prone. Support engineers handling multiple incidents may overlook approaching SLA deadlines, while managers often lack real-time visibility into incidents that are at risk of breaching their SLAs. Consequently, organizations spend additional effort investigating SLA violations after they occur instead of preventing them proactively.

To address these challenges, this project proposes a **Virtual Agent–Driven SLA Breach Awareness & Justification System** developed using the ServiceNow platform. The solution combines Incident Management, SLA Management, Flow Designer, Virtual Agent, Notifications, Reports, and Platform Analytics to automate SLA governance. The system continuously monitors SLA status, detects incidents approaching breach conditions, sends proactive notifications to assignees, collects mandatory breach justifications through an interactive Virtual Agent, and provides analytical dashboards for managers to monitor SLA compliance effectively.

The proposed solution is implemented entirely using native ServiceNow capabilities without requiring complex custom coding. This makes the application scalable, maintainable, and suitable for organizations seeking a low-code approach to improve SLA governance. By integrating automation, conversational AI, and real-time analytics, the project demonstrates how ServiceNow can be leveraged to improve service quality while reducing manual effort.

1.2 Purpose

The primary purpose of this project is to develop an automated governance solution that minimizes SLA breaches and improves accountability within IT service operations. Instead of

relying on manual monitoring, the system proactively identifies incidents approaching SLA breach conditions and immediately notifies responsible users before violations occur. Early awareness enables support engineers to take corrective actions and prioritize critical incidents, thereby improving overall service quality.

Another objective of the project is to enforce proper documentation of SLA breaches. Whenever an SLA violation occurs, the system ensures that users provide a mandatory breach justification through the ServiceNow Virtual Agent. This creates standardized records that improve transparency, simplify audits, and help management analyze recurring operational issues.

The major objectives of the project are:

- Automate SLA risk monitoring.
- Notify users before SLA breaches occur.
- Collect mandatory breach justifications.
- Improve SLA governance and accountability.
- Reduce manual monitoring activities.
- Provide real-time reports and dashboards.
- Enhance customer satisfaction through timely incident resolution.

Overall, the project demonstrates how automation and conversational AI can significantly improve ITSM processes while ensuring better SLA compliance.

2. IDEATION PHASE

2.1 Problem Statement

Every organization providing IT services is expected to resolve customer issues within predefined Service Level Agreements (SLAs). These agreements specify the maximum allowable response and resolution times for incidents based on their priority. Failure to meet these commitments results in SLA breaches, which negatively affect customer trust, business reputation, and operational efficiency.

In many organizations, SLA breaches occur not because employees lack technical expertise but because they are unaware that an SLA deadline is approaching. Support engineers typically manage numerous incidents simultaneously, making it difficult to continuously monitor SLA timers manually. Managers also face challenges in identifying which incidents require immediate attention since traditional monitoring methods provide limited proactive visibility.

Another major issue is the absence of standardized breach documentation. When an SLA is violated, users often fail to provide proper reasons or explanations for the delay. Missing

justifications reduce accountability, complicate audits, and make it difficult for management to identify recurring operational bottlenecks.

Existing systems primarily focus on reporting SLA breaches after they occur rather than preventing them. They provide limited user interaction and lack intelligent guidance that encourages timely action. Consequently, organizations spend considerable effort resolving

2.2 Empathy Map Canvas

Understanding the needs and challenges of users is one of the most important phases of designing an effective solution. The Empathy Map Canvas was used to study the daily activities, expectations, frustrations, and motivations of the stakeholders involved in SLA management. The primary users identified for this project are **IT Support Engineers**, **Incident Managers**, and **Service Delivery Managers** who interact with incidents throughout their lifecycle.

The empathy study also revealed that users prefer simple and guided workflows rather than navigating multiple ServiceNow forms. They expect timely reminders, automated notifications, conversational guidance, and centralized reporting that minimizes manual effort. These observations strongly supported the development of an intelligent Virtual Agent capable of proactively interacting with users whenever an SLA is at risk.

The insights collected through the Empathy Map became the foundation for designing a user-centric solution that focuses on automation, accountability, ease of use, and proactive governance. By understanding what users think, feel, say, and do during their daily work, the proposed system effectively addresses real operational challenges instead of merely providing additional reporting functionality.

Empathy Map Summary

Pain Points

- Manual SLA monitoring.
- Missed deadlines.
- Missing breach justifications.
- Limited visibility.

Gain Points

- Automatic reminders.
- Easy acknowledgement.
- Guided justification.
- Real-time dashboards.

- **2.3 Brainstorming**

The brainstorming phase focused on identifying practical solutions for improving SLA governance while minimizing manual effort. Initially, several approaches were discussed based on existing IT service management practices. The objective was to determine a solution that could proactively reduce SLA breaches, improve communication between stakeholders, and provide management with meaningful operational insights.

Different ideas were explored, including periodic reminder emails, manual approval workflows, dashboard-only monitoring systems, scheduled notifications, escalation mechanisms, AI chatbots, and automated reporting solutions. Each idea was evaluated based on implementation complexity, user experience, automation capability, scalability, and compatibility with the ServiceNow platform.

The integration of ServiceNow Virtual Agent with Flow Designer emerged as the most effective solution because it combines proactive notifications, conversational interaction, mandatory justification collection, and centralized reporting within a single platform. The Virtual Agent provides an intuitive interface that guides users through SLA acknowledgement while automatically recording governance information in the incident record.

Brainstorming Process

Step 1 – Identify the Problem

- Frequent SLA breaches.
- Manual monitoring.
- Missing justifications.
- Limited manager visibility.

Step 2 – Generate Ideas

- Reminder Emails
- Dashboard Monitoring
- Scheduled Notifications
- Virtual Agent
- Reports & Analytics

Step 3 – Group Similar Ideas

- Notification Solutions
- Automation Solutions

- Reporting Solutions
- Governance Solutions

Step 4 – Prioritize Ideas

Evaluation Criteria:

- Easy to implement
- Low maintenance
- High automation
- Better user experience
- ServiceNow compatibility

Final Selected Idea

Virtual Agent–Driven SLA Breach Awareness & Justification System

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The Customer Journey Map illustrates the complete interaction between users and the proposed system from incident creation until SLA completion. Mapping this journey helps identify important touchpoints where automation can improve efficiency and reduce manual effort. The customer journey also ensures that every stage of the SLA lifecycle contributes toward better governance and enhanced service quality.

The process begins when a support engineer creates or receives an incident. Immediately after incident creation, the appropriate SLA is attached based on incident priority and business rules. The ServiceNow platform continuously monitors the SLA timer throughout the incident lifecycle. When the remaining SLA time reaches the predefined risk threshold, Flow Designer automatically detects the condition and initiates the SLA governance workflow.

The assigned support engineer receives a notification informing them that the incident is approaching an SLA breach. Simultaneously, the Virtual Agent initiates a conversation requesting the user to acknowledge the SLA risk. If the incident eventually breaches the SLA, the Virtual Agent prompts the user to provide a mandatory breach reason and justification. The submitted information is stored within the incident record for future reference and auditing.

Managers receive automated notifications and can monitor the entire process through reports and dashboards that display SLA trends, breach statistics, acknowledgement status,

assignment group performance, and pending justifications. This complete journey improves accountability while reducing the dependence on manual supervision.

Customer Journey Steps

1. Incident Created.
2. SLA Automatically Attached.
3. SLA Timer Starts.
4. System Monitors SLA Continuously.
5. SLA Approaches Risk Threshold.
6. Flow Designer Triggered.
7. Email Notification Sent.
8. Virtual Agent Starts Conversation.
9. User Acknowledges SLA Risk.
10. User Provides Breach Justification.
11. Incident Record Updated.
12. Reports Generated.
13. Dashboard Updated.
14. Manager Reviews Performance.
15. Incident Closed.

3.2 Solution Requirements

Developing an effective SLA governance system requires identifying both the functional capabilities and quality attributes expected from the application. Solution requirements define the features that the system must provide to satisfy business objectives while ensuring reliability, security, usability, and scalability. The proposed solution has been designed using ServiceNow's native capabilities so that organizations can implement the application with minimal customization while achieving maximum automation.

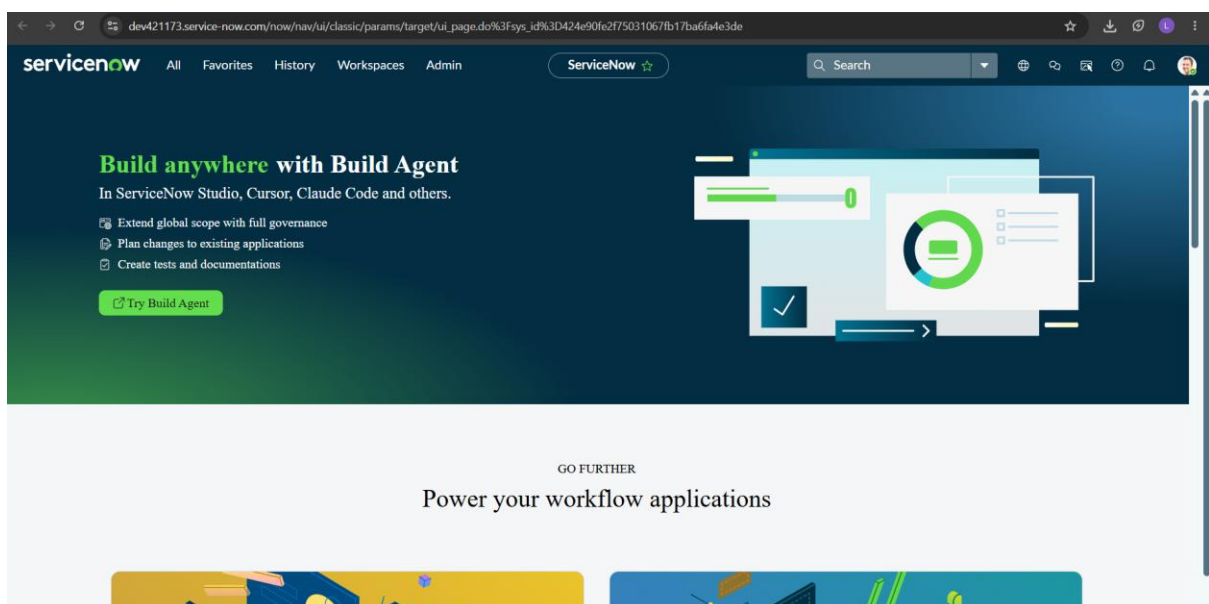
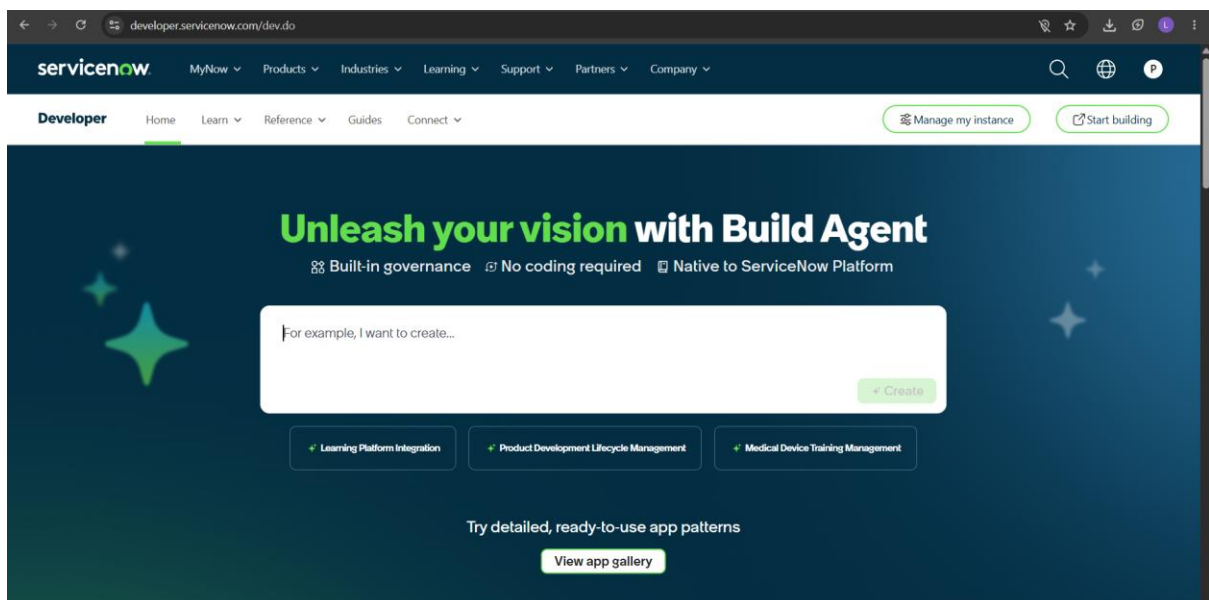
The primary objective of the solution is to reduce SLA breaches through proactive monitoring rather than reactive reporting. The application continuously evaluates SLA status, automatically identifies incidents approaching SLA breach conditions, and initiates predefined workflows whenever SLA thresholds are reached. These workflows reduce manual intervention while improving communication between support engineers and management.

Another important requirement is improving accountability. Whenever an SLA breach occurs, the system ensures that users acknowledge the risk and provide mandatory breach justifications before closing the incident. This information is permanently stored within ServiceNow records, supporting compliance, audit readiness, and operational analysis.

The solution must also provide managers with comprehensive reporting capabilities. Analytical dashboards display SLA breach trends, assignment group performance, acknowledgement status, pending justifications, and recurring breach reasons. These reports enable management to make informed decisions for improving IT service delivery.

Functional Requirements

The functional requirements describe the core features implemented in the proposed system.



- FR-1:** Automatically monitor SLA status for every incident.
- FR-2:** Detect incidents approaching SLA breach conditions.
- FR-3:** Trigger Flow Designer workflows when SLA becomes "At Risk."
- FR-4:** Send automated email notifications to assigned users and managers.
- FR-5:** Launch Virtual Agent conversations automatically.
- FR-6:** Collect mandatory SLA breach acknowledgements.
- FR-7:** Store breach justifications within Incident records.
- FR-8:** Generate SLA reports and dashboard analytics.
- FR-9:** Display pending breach justifications.
- FR-10:** Support centralized SLA governance.

Non-Functional Requirements

The non-functional requirements ensure the quality and reliability of the application.

- NFR-1:** User-friendly interface.
- NFR-2:** Secure role-based access.
- NFR-3:** High system availability.
- NFR-4:** Fast workflow execution.
- NFR-5:** Scalable architecture.
- NFR-6:** Easy maintenance.

3.3 Data Flow Diagram

A Data Flow Diagram (DFD) provides a graphical representation of how information moves throughout the proposed system. It identifies the major entities, processes, data stores, and information exchanged during SLA monitoring and governance. The DFD serves as a blueprint that explains how the system processes data from the moment an incident is created until SLA reports are generated.

The proposed solution begins with the creation of an incident by an IT support engineer. Once the incident is submitted, ServiceNow automatically associates the corresponding SLA based on predefined business rules. The SLA timer starts immediately and continues running throughout the incident lifecycle.

The SLA monitoring component continuously evaluates the remaining SLA duration. When the remaining time reaches the configured threshold, Flow Designer automatically detects the condition and triggers the notification workflow. The assigned engineer receives an email notification while the Virtual Agent simultaneously starts an interactive conversation requesting SLA acknowledgement.

If the SLA eventually breaches, the Virtual Agent requests a mandatory breach justification from the user. After submission, the justification is validated and stored inside the Incident record. Managers receive email notifications while Platform Analytics updates reports and dashboards using the latest governance information.

The DFD demonstrates that information flows automatically between ServiceNow modules without requiring continuous manual intervention. This significantly improves operational efficiency while ensuring better SLA compliance.

Data Flow Process

Step 1: User creates an Incident.

Step 2: Incident is stored in ServiceNow.

Step 3: SLA is attached automatically.

Step 4: SLA timer starts.

Step 5: System continuously monitors SLA.

Step 6: Flow Designer detects SLA risk.

Step 7: Email notification is sent.

Step 8: Virtual Agent starts conversation.

Step 9: User acknowledges SLA risk.

Step 10: User submits breach justification.

Step 11: Justification is stored.

Step 12: Reports are generated.

Step 13: Dashboard is updated.

Step 14: Manager reviews SLA performance.

Step 15: Incident is resolved.

3.4 Technology Stack

The proposed solution has been developed entirely using the ServiceNow platform, allowing seamless integration between IT Service Management components and automation capabilities. The selected technology stack follows a low-code architecture that reduces development complexity while providing enterprise-level scalability, security, and maintainability.

ServiceNow Incident Management serves as the central application for managing support requests. SLA Management continuously monitors response and resolution commitments. Flow Designer automates business processes without requiring custom programming. The Virtual Agent enables conversational interactions that simplify user acknowledgement and justification activities. Platform Analytics provides real-time dashboards and reports that support operational decision-making.

The cloud-based nature of ServiceNow ensures high availability and secure access from anywhere. Since all major components belong to the same ecosystem, data synchronization remains consistent while minimizing integration challenges.

Technology Components

Component	Purpose
ServiceNow Platform	Application Development
Incident Management	Ticket Management
SLA Management	SLA Monitoring
Flow Designer	Workflow Automation
Virtual Agent	User Interaction
UI Policies	Mandatory Field Validation
Notifications	Email Alerts
Reports	SLA Analytics
Dashboards	Performance Monitoring
Platform Analytics	Business Intelligence
ServiceNow Database	Data Storage

4. PROJECT DESIGN

4.1 Problem–Solution Fit

Selecting an appropriate solution requires understanding how effectively the proposed application addresses the identified business problems. The Problem–Solution Fit analysis confirms that the Virtual Agent–Driven SLA Breach Awareness & Justification System directly resolves the challenges identified during the ideation and requirement analysis phases.

The major problems identified include manual SLA monitoring, delayed notifications, missing breach justifications, limited managerial visibility, and insufficient operational accountability. Each of these issues contributes to increasing SLA violations and reducing customer satisfaction.

The proposed solution introduces automated monitoring, intelligent notifications, conversational acknowledgement, mandatory justification, centralized reporting, and dashboard visualization. These capabilities collectively improve operational efficiency while reducing manual intervention.

Unlike conventional SLA monitoring systems, the proposed application actively communicates with users before SLA breaches occur. Managers receive analytical insights that enable preventive decision-making instead of reacting after violations have already occurred. This strong alignment between identified problems and implemented features confirms that the proposed solution satisfies both technical and business requirements.

Problem–Solution Mapping

Problem	Proposed Solution
Manual SLA monitoring	Automated SLA monitoring
Missed SLA deadlines	Early SLA notifications
No acknowledgement	Virtual Agent interaction
Missing justifications	Mandatory justification collection
Limited visibility	Reports & Dashboards
Poor accountability	Centralized SLA governance

4.2 Proposed Solution

The proposed solution, **Virtual Agent–Driven SLA Breach Awareness & Justification System**, is designed to improve SLA governance by automating the monitoring, notification, acknowledgement, and reporting processes within the ServiceNow platform. Instead of relying on manual supervision, the system continuously evaluates SLA status and

automatically initiates workflows whenever an incident approaches a breach condition. This proactive approach minimizes delays, improves accountability, and enhances the overall efficiency of IT service management.

The solution integrates multiple ServiceNow modules including Incident Management, SLA Management, Flow Designer, Virtual Agent, Notifications, Reports, and Platform Analytics. These components work together to provide an end-to-end automated governance mechanism. Whenever an incident is created, the appropriate SLA is attached automatically. The system continuously monitors the SLA timer and evaluates predefined risk thresholds.

As soon as the SLA reaches an "At Risk" condition, Flow Designer automatically triggers the workflow. The assigned support engineer immediately receives an email notification informing them about the approaching SLA breach. Simultaneously, the Virtual Agent starts an interactive conversation that guides the user through acknowledging the SLA risk. If the SLA is breached, the Virtual Agent requests a mandatory breach justification before the incident can proceed further.

The collected information is securely stored within the Incident record, allowing managers to review historical data for audits and performance analysis. Reports and dashboards automatically display key performance indicators including breach reasons, acknowledgement status, pending justifications, assignment group performance, and SLA trends. These analytical insights help management identify operational bottlenecks and improve future service delivery.

The proposed solution minimizes manual intervention, standardizes SLA governance, improves communication between stakeholders, and provides organizations with a scalable, low-code solution that can be implemented without extensive software development.

Proposed Solution Workflow

Step 1: Incident is created.

Step 2: SLA is automatically attached.

Step 3: SLA monitoring begins.

Step 4: Flow Designer continuously checks SLA status.

Step 5: SLA reaches the risk threshold.

Step 6: Email notification is sent to the assignee.

Step 7: Virtual Agent starts automatically.

Step 8: User acknowledges the SLA risk.

Step 9: User submits breach justification.

Step 10: Justification is stored.

Step 11: Reports and dashboards are updated.

Step 12: Manager monitors SLA compliance.

4.3 Solution Architecture

The architecture of the proposed system follows a modular and centralized design using native ServiceNow components. Each module performs a specific responsibility while working together to achieve complete SLA governance. The architecture ensures that information flows seamlessly between different components, enabling automation without requiring complex custom programming.

At the user level, support engineers interact with Incident Management for creating, updating, and resolving incidents. Incident records are automatically linked with Service Level Agreements according to predefined business rules. SLA Management continuously monitors the remaining response and resolution time associated with each incident.

Whenever an SLA approaches the configured threshold, Flow Designer immediately triggers the automation process. Notifications are sent to the responsible assignee, and the Virtual Agent begins an interactive conversation requesting acknowledgement and breach justification. The responses collected from users are validated through UI Policies and stored in ServiceNow tables.

Managers receive updated information through Platform Analytics, which automatically generates reports and dashboards. These dashboards provide centralized visibility into SLA performance, breach reasons, pending justifications, assignment group statistics, and overall governance metrics. Since all modules belong to the same ServiceNow ecosystem, the architecture remains secure, scalable, and easy to maintain.

The architecture supports future enhancements such as predictive SLA analysis, AI-driven recommendations, mobile notifications, and advanced analytics without requiring significant modifications to the existing system.

Architecture Components

Layer	Description
User Layer	IT Support Engineers and Managers interact with the application.
Incident Layer	Stores and manages Incident records.
SLA Layer	Monitors SLA timers continuously.
Automation Layer	Flow Designer executes automated workflows.
Interaction Layer	Virtual Agent communicates with users.

Validation Layer	UI Policies enforce mandatory justifications.
Notification Layer	Sends Email Notifications.
Database Layer	Stores incidents and governance information.
Analytics Layer	Reports and Dashboards display SLA metrics.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Effective project planning is essential for ensuring that software development activities are completed within the planned schedule while maintaining quality. The proposed project follows an Agile-inspired development approach in which the implementation work is divided into multiple sprints. Each sprint focuses on delivering a specific set of functionalities, allowing continuous progress monitoring and incremental improvements throughout the project lifecycle.

The planning process began by identifying all functional requirements and converting them into user stories. These user stories were prioritized according to business value, implementation dependency, and project objectives. Story points were assigned based on estimated development effort, enabling effective sprint planning and workload distribution.

Sprint 1 focused on the foundation of the application by implementing Incident Management, SLA attachment, and automated SLA monitoring. Sprint 2 concentrated on Flow Designer automation, notification workflows, and Virtual Agent conversations. Sprint 3 focused on reporting, dashboards, and governance analytics. Sprint 4 was dedicated to testing, validation, defect resolution, and deployment.

This structured planning approach ensured that each development milestone was completed systematically while allowing testing and feedback after every sprint. The incremental methodology reduced project risks and improved overall development quality.

Sprint Planning

Sprint 1

Objective: Build the SLA monitoring foundation.

Activities:

- Configure Incident Management.
- Configure SLA Definitions.
- Attach SLA automatically.

- Verify SLA timer.

Deliverables:

- Incident Creation
- SLA Running
- SLA Monitoring

Sprint 2

Objective: Automate SLA governance.

Activities:

- Configure Flow Designer.
- Configure Notifications.
- Develop Virtual Agent Topic.
- Configure UI Policies.

Deliverables:

- SLA At Risk Detection
- Email Notifications
- Virtual Agent
- Justification Collection

Sprint 3

Objective: Develop reporting components.

Activities:

- Create Reports.
- Configure Dashboard.
- Generate Analytics.
- Validate Report Filters.

Deliverables:

- SLA Reports

- Dashboard
- Platform Analytics

Sprint 4

Objective: Validate and deploy.

Activities:

- Functional Testing.
- Performance Testing.
- User Acceptance Testing.
- Bug Fixing.
- Final Deployment.

Deliverables:

- Tested Application
- Bug-Free Solution
- Final Project

Benefits of Proper Project Planning

- Improved task management.
- Better resource utilization.
- Clear sprint objectives.
- Reduced project risks.
- Faster issue identification.
- Timely completion of milestones.
- Better quality assurance.
- Easier progress tracking.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Performance testing was conducted to evaluate the efficiency, reliability, and responsiveness of the proposed **Virtual Agent–Driven SLA Breach Awareness & Justification System** under normal operating conditions. The primary objective of this phase was to ensure that every functional component performs accurately while maintaining acceptable response times and consistent system behavior.

The testing process included validating Incident Management, SLA attachment, SLA monitoring, Flow Designer automation, Email Notifications, Virtual Agent conversations, UI Policies, Reports, and Dashboard functionalities. Each module was tested independently before performing integrated system testing. This approach ensured that individual components functioned correctly while maintaining smooth interaction with other ServiceNow modules.

The system demonstrated consistent performance throughout the testing process. SLA monitoring was performed automatically without affecting incident processing. Flow Designer executed workflows successfully whenever SLA risk conditions were detected. Email notifications reached the assigned users promptly, and Virtual Agent conversations were initiated automatically according to the configured trigger conditions. Reports and dashboards displayed updated SLA information after every workflow execution, enabling managers to monitor SLA performance without delays.

6.2 Functional Testing

Functional testing was performed to verify that every feature of the proposed system operates according to the specified business requirements. Each module was tested individually and then validated as part of the complete SLA governance workflow. The testing ensured that every automation, notification, user interaction, and reporting function produced the expected results.

The testing process started with Incident Creation and SLA attachment. After confirming that SLA timers were attached successfully, SLA risk detection was validated by simulating incidents approaching predefined SLA thresholds. Flow Designer correctly identified SLA risk conditions and initiated automated workflows without manual intervention.

Email notifications were verified to ensure that assigned support engineers received timely alerts regarding approaching SLA breaches. The Virtual Agent was tested to confirm that users could acknowledge SLA risks and submit mandatory breach justifications through conversational interaction. UI Policies were validated to ensure that justification fields became mandatory whenever SLA risk conditions occurred.

Finally, reports and dashboards were tested to verify the accuracy of SLA breach statistics, acknowledgement status, pending justifications, and assignment group analytics. Every functional requirement specified during requirement analysis was successfully validated through structured test cases.

A total of **11 functional test cases** were executed to validate the proposed solution.

Test Case	Result
Incident Creation	Pass
SLA Attachment & Running	Pass
SLA At Risk Trigger	Pass
SLA Notification Email	Pass
UI Policy Validation	Pass
Virtual Agent Invocation	Pass
SLA Acknowledgement	Pass
SLA Breaches by Reason Report	Pass
Acknowledged vs Unacknowledged Report	Pass
Pending Justifications Report	Pass
SLA Governance Dashboard	Pass

All test cases passed successfully without major functional defects.

6.5 Defect Analysis

During the initial stages of testing, a few minor configuration-related issues were identified. These included dashboard widget alignment, report filter configuration, Virtual Agent topic mapping, and notification timing. Since these issues were related to ServiceNow configuration rather than application logic, they were resolved through administrative configuration updates.

After implementing the necessary corrections, all affected modules were retested. The second round of testing confirmed that the defects had been completely resolved. No critical or high-severity defects remained in the application. The final version of the solution demonstrated stable functionality across all implemented features.

7.Results:

7.1 Process Screenshots

The implementation of the Virtual Agent–Driven SLA Breach Awareness & Justification System successfully achieved the intended project objectives. Screenshots collected during implementation demonstrate the complete workflow from incident creation to dashboard visualization. Each stage of the application executed successfully according to the designed architecture and workflow.

The screenshots includes the process we implemented while implementing this project:

The image shows two screenshots from the ServiceNow interface. The top screenshot displays the 'Tables' list, and the bottom screenshot shows the 'Dictionary Entry' form for a new record.

Tables List:

Label	Name	Extends table	Extensible	Updated
Search	Search	Search	Search	Search
Incident	incident	Task	false	2026-04-30 01:05:41
Incident Fact Table	incident_fact_table	(empty)	false	2026-04-30 00:16:50
Incident Task	incident_task	Task	false	2026-04-30 00:38:14
Workspace Session Tab Setting	information_session_tab_workspace_setting	(empty)	false	2026-04-30 01:15:31
ServiceNow Instance	instance	(empty)	false	2026-04-30 00:19:29
Instance Subscription Entitlement	instance_subscription_entitlement	(empty)	false	2026-04-30 01:10:41
Allowed Tables	instance_toggle_allowed_tables	(empty)	false	2026-04-30 01:20:11
Not Allowed Tables	instance_toggle_notallowed_tables	(empty)	false	2026-04-30 01:20:11
Interaction	interaction	(empty)	true	2026-04-30 00:54:56
Interaction Context	interaction_context	(empty)	false	2026-04-30 00:54:56
Interaction Help Request	interaction_help_request	(empty)	true	2026-04-30 00:54:55
Interaction Help Request Configuration	interaction_help_request_configuration	Application File	false	2026-04-30 00:54:54
Interaction Help Request Reason	interaction_help_request_reason	Application File	false	2026-04-30 00:54:56
JSON Blob	interaction_json_blob	(empty)	false	2026-04-30 00:54:56
Interaction Log	interaction_log	(empty)	false	2026-04-30 00:54:54
Interaction Related Record	interaction_related_record	(empty)	false	2026-04-30 00:54:55

Dictionary Entry Form:

Table: Incident [incident]

Type: Choice

Column label: SLA Breach Reason

Column name: u_sla_breach_reason

Max length:

Application: Global

Active: ☒

Function field: ☐

Mandatory: ☐

Display: ☐

Read only option: -- None --

List Specification:

is a list of suggested values in a **Choice** list. In the Advanced view you can select the **Choice** table and the **Choice** field to take choice values from, plus a **Dependent** field.

Choice: -- None --

Links: [view](#)

dev421173.service-now.com/now/nav/ui/classic/params/target/sys_dictionary.do%3Fsys_id%3D5d311862f1871067fb17ba6fa4e319%26sysparm_record_list%3DnameONE%2BINcident%252Ctask%2...

service-nowAllFavoritesHistoryWorkspacesDictionary Entry - SLA Breach ReasonSearchCreate Choice ListDelete ColumnUpdate

Dictionary EntrySLA Breach ReasonCreate Choice ListDelete ColumnUpdate

Related LinksShow TableRun Point ScanAdvanced view

List controlsrolsChoices (6)AttributesLabels (1)

LabelSearch

Actions on selected rows...New

Label	Value	Language	Sequence	Inactive	Updated
Dependency on third party	1	en		1 false	2026-07-06 04:40:41
Incorrect categorization	3	en		3 false	2026-07-06 04:43:17
System outage	4	en		4 false	2026-07-06 04:46:29
Awaiting user response	2	en		2 false	2026-07-06 04:42:01
Other	6	en		6 false	2026-07-06 04:48:02
Approval delay	5	en		5 false	2026-07-06 04:47:20
Insert a new row...					

1 to 6 of 6

dev421173.service-now.com/now/app-manager/home/app/id/134450a0b378130078162ac16a8dc1e/v/9.3.2

service-nowAllFavoritesHistoryWorkspacesAdminApplication ManagerSearch

Application managerITSM Virtual Agent ConversationsActivity logRequest pluginOpen store

ITSM Virtual Agent Conversations

App Id: sn_itsm_vaLatest Version: 9.3.2 | Industry: All | Category:

Summary

Get instant resolution to repetitive IT service tasks and reduce call volumes with a 24/7 self-service expert chatbot that understands natural human language. Customers and employees with a 24/7 self-service expert chatbot that understands natural human language. Customers and employees with a 24/7 self-service expert chatbot that understands natural human language.

Show more

Pricing Paid

Details

Version: 9.3.2 (Latest)

Compatibility and Impact

Installation progress

ITSM Virtual Agent Conversations

Install from the App RepositoryRunning 91%

Executing queued operation

Installing application packageRunning 91%

Run in background

Helpful Resources

[Product Brief](#)

[ServiceNow Store Terms Of Use](#)

[ITSM Virtual Agent do...](#)

[ITSM Virtual Agent Co...](#)

Basic properties

* Application scope ⓘ

Global



* Type

Topic



* Internal name ⓘ

sla_breach_awareness_justification

* Display name ⓘ

SLA Breach Awareness & Justification

Topic description

Describe the purpose and functionality of this topic

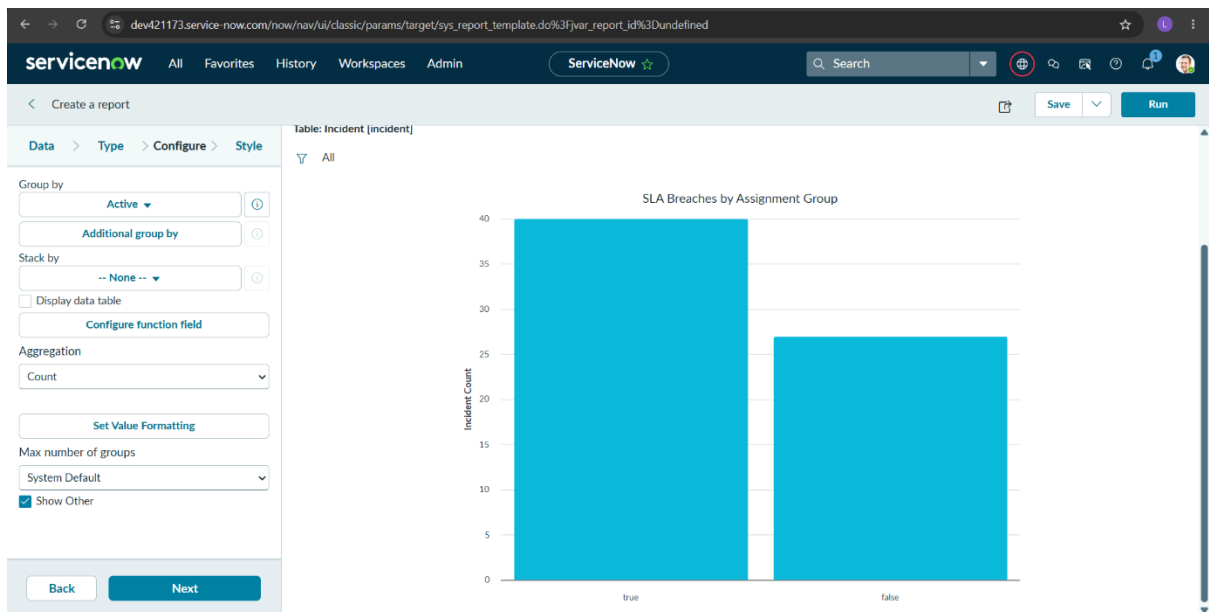
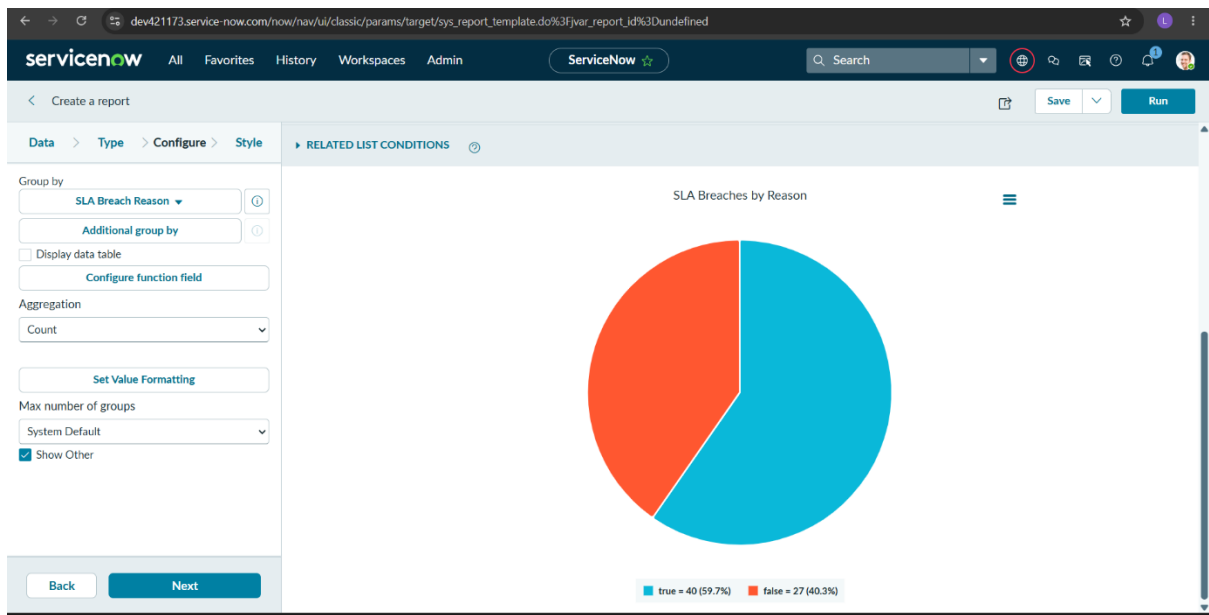


The screenshot displays the ServiceNow Virtual Agent Designer interface. The top navigation bar includes the ServiceNow logo, user profile, and search bar. The main header shows the current workspace: "Virtual Agent Designer" and the topic name "sla_breach_awareness_justification".

The interface is divided into several sections:

- Components Panel (Left):** Lists various building blocks categorized under "Components" and "Utilities". Categories include HTML, Multi-res..., Script, Card, Table, Video, Lookup, Script Action, Record Action, Decision, Topic Block, Action, Custom Control, Deflection, Diagnostic Topic..., Geolocation, Multi, and Pause.
- Flowchart (Center):** A visual representation of the conversation flow. It starts with a "Start" node, followed by "Greeting", "Manage your SLA", and "Acknowledge SLA Plan". A decision node labeled "Decision" branches into two paths: "Yes" leading to "Send Slack Tip" and "No" leading to "Decision to Fail".
- Properties Panel (Right):** Currently shows the "Decision utility" configuration. It includes a description: "Add two or more branches that represent different conversation paths. [Learn more](#)". The "Node name" field is set to "Decision".

The bottom status bar indicates the application is set to "Global".



dev421173.service-now.com/now/nav/ui/classic/params/target/sys_report_template.do%3Fjvar_report_id%3Dundefined

ServiceNow

Create a report

Save Run

Data > Type > Configure > Style

Choose columns

Group by

-- None --

Additional group by

Configure function field

Table: Incident [incident]

All

Add Sort

INCIDENT C

All of these con

SLA at Risk

SLA Breach

or

New Criteria

RELATED LIST CONDITIONS

Back Next

Pending Justifications

Columns

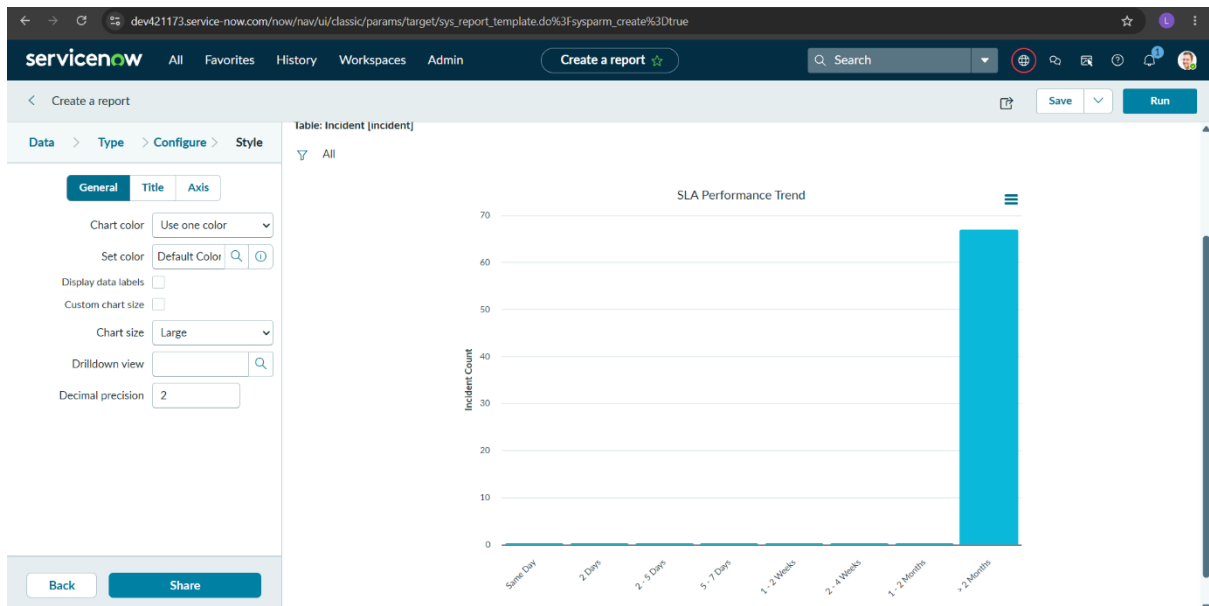
Available

- Assignment group [+]
- Assigned to [+]
- Active
- Activity dum
- Actual end
- Actual start
- Additional assignee list
- Approval
- Approval history
- Approval set
- Business duration
- Business impact
- Business resolve time
- Caller [+]
- Category

Selected

- Number
- Assignment group
- Assigned to
- SLA dum

OK



7.2 Output:

The implementation results demonstrate the successful execution of the proposed SLA governance solution. Screenshots captured during testing illustrate each stage of the workflow, including incident creation, SLA attachment, risk detection, notification delivery, Virtual Agent acknowledgement, report generation, and dashboard visualization. These outputs confirm that the developed solution meets the intended functional objectives and operates as expected within the ServiceNow platform.

8. ADVANTAGES & DISADVANTAGES

Advantages

The proposed system provides several benefits that improve IT service management and SLA governance.

- Proactive SLA monitoring reduces the likelihood of SLA breaches.
- Automated notifications improve user awareness.
- Virtual Agent provides interactive user guidance.
- Mandatory breach justification improves accountability.
- Flow Designer eliminates repetitive manual activities.
- Reports provide meaningful analytical insights.
- Dashboards enable centralized SLA monitoring.
- Native ServiceNow implementation minimizes development complexity.
- Scalable design supports organizational growth.

Disadvantages

Although the proposed solution provides significant improvements, certain limitations remain.

- Proper SLA configuration is required before deployment.
- Organizations must configure Virtual Agent topics.
- Notification effectiveness depends on email availability.
- Dashboard data accuracy depends on correct incident updates.
- Initial implementation requires ServiceNow administrative knowledge.
- The solution currently focuses only on Incident Management and can be extended to other ITSM modules.

9. CONCLUSION

The **Virtual Agent–Driven SLA Breach Awareness & Justification System** successfully demonstrates how automation can improve Service Level Agreement governance within IT service management. By integrating ServiceNow Incident Management, SLA Management, Flow Designer, Virtual Agent, Notifications, Reports, and Platform Analytics, the proposed solution transforms traditional manual SLA monitoring into a proactive and intelligent governance process.

The implementation achieved all project objectives by automatically identifying SLA risks, notifying responsible users before SLA breaches occur, collecting mandatory breach justifications through conversational interaction, and providing managers with real-time dashboards for monitoring organizational SLA performance. The application significantly reduces manual effort while improving accountability, transparency, and operational efficiency.

Testing results confirmed that all functional requirements were successfully implemented. Every workflow executed according to the designed architecture, and all defined test cases passed successfully during functional and user acceptance testing. The system proved reliable, scalable, and suitable for deployment in enterprise IT environments.

The project demonstrates the effectiveness of low-code development using the ServiceNow platform while highlighting the value of intelligent automation in improving service quality. Future enhancements may include AI-based SLA prediction, mobile notifications, predictive analytics, machine learning recommendations, and integration with additional ITSM modules such as Problem Management, Change Management, and Request Management.

Overall, the proposed solution represents a practical, scalable, and user-friendly approach to modern SLA governance that enhances customer satisfaction, strengthens operational compliance, and supports continuous improvement in IT service delivery.